



DATA STRUCTURE AND ALGORITHM

23CSE203

LAB MANUAL

SUBMITTED BY

NAME	G. HEMA HARSHINI
ROLL NO	AV.SC.U4CSE25108
YEAR/SEM/SECTION	2 ST YEAR/SEM-3/CSE-B
CREDITS	05

SUBMITTED TO

NAME	DR.RAJ KUMAR BATCHU
DESIGNATION	ASST.PROFFESSOR(S1. Grade)
DEPARTMENT	SCHOOL OF COMPUTING

INDEX

S.NO	PROGRAM	PAGE NO	REMARKS
1.	WEEK 1: 1a) A rocket launch displays a countdown before launch. Instead of using a loop, implement the countdown using recursion. Write a recursion python program to display a countdown from n to 1, followed by "Launch!". 1b) Banks calculate the growth of an investment using compound interest. The formula requires calculating powers, which can be implemented recursively, write a recursive program to calculate p^n (power) where: p =principal growth factor, n =no. of years. 1c) An HR department stores employee IDs in a list. Develop a recursive program to search for a given employee ID. Write a recursive function search for an employee ID in a list. 1d) In a logistics company, the no of possible ways to arrange n different parcels are calculated using the factorial formula($n!$). Develop a recursive program to calculate the factorial of a given number. 1e) Write a recursive python program to print the 1 st 'n' terms of Fibonacci series.		
2.	WEEK 2: 2a) Write a python program to implement linear search. 2b) (i) Write a python program to implement binary search with sorted array as an input. (ii) Write a python program to implement binary search without sorted array as an input.		

WEEK-1

1 a) **AIM:** A rocket launch displays a countdown before launch. Instead of using a loop, implement the countdown using recursion. Write a recursion python program to display a countdown from n to 1, followed by “Launch!”.

CODE:

```
def countdown(n):
```

```
    if n>=1:
```

```
        print(n)
```

```
        countdown(n-1)
```

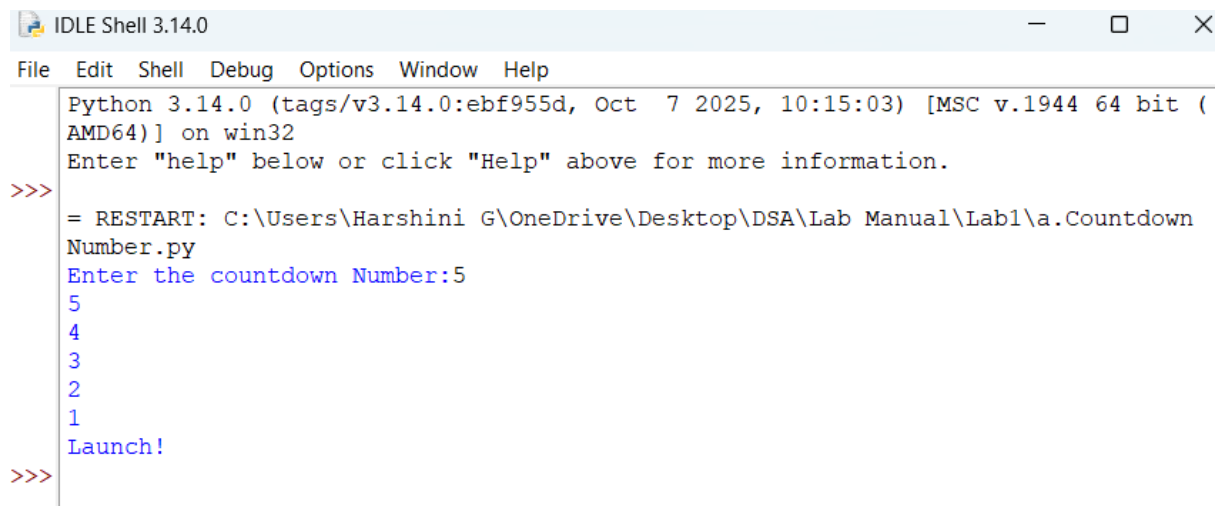
```
    else:
```

```
        print("Launch!")
```

```
n=int(input("Enter the countdown Number:"))
```

```
countdown(n)
```

OUTPUT:



```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:\Users\Harshini G\OneDrive\Desktop\DSA\Lab Manual\Lab1\a.Countdown
Number.py
Enter the countdown Number:5
5
4
3
2
1
Launch!
>>>
```

1 b) **AIM:** Banks calculate the growth of an investment using compound interest. The formula requires calculating powers, which can be implemented recursively, write a recursive program to calculate p^n (power) where:

p=principal growth factor, n=no. of years.

CODE:

```
if n<=0:
```

```

        return 1

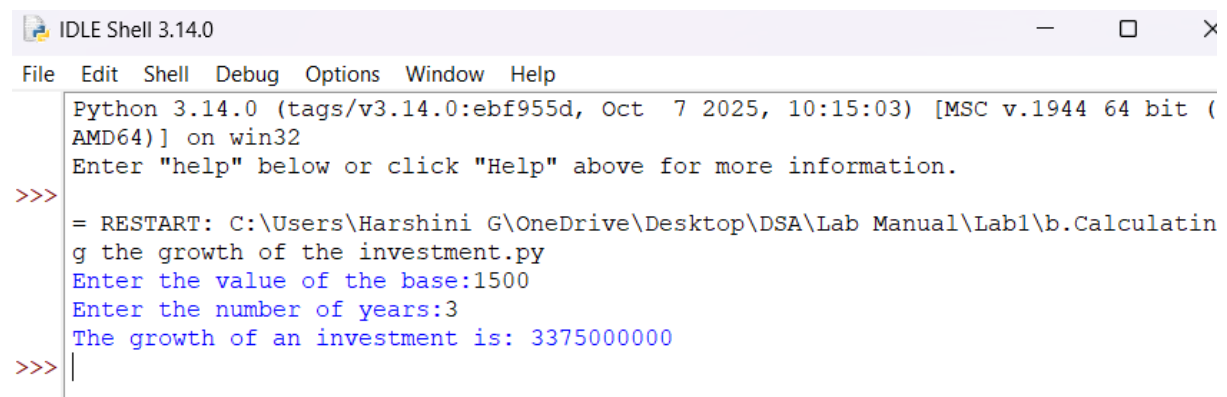
    else:

        return p*power(p,n-1)

p=int(input("Enter the value of the base:"))
n=int(input("Enter the number of years:"))
result=power(p,n)
print("The growth of an investment is:",result)

```

OUTPUT:



```

IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:\Users\Harshini G\OneDrive\Desktop\DSA\Lab Manual\Lab1\b.Calculating the growth of the investment.py
Enter the value of the base:1500
Enter the number of years:3
The growth of an investment is: 3375000000
>>> |

```

1c) **AIM:** An HR department stores employee IDs in a list. Develop a recursive program to search for a given employee ID. Write a recursive function search for an employee ID in a list.

CODE:

```

def binary_search(arr,key):

    low=0

    high=len(arr)-1

    while low<=high:

        mid=(low+high)//2

        if arr[mid]==key:

            return mid

        elif arr[mid]<key:

            low=mid+1

        else:

```

```

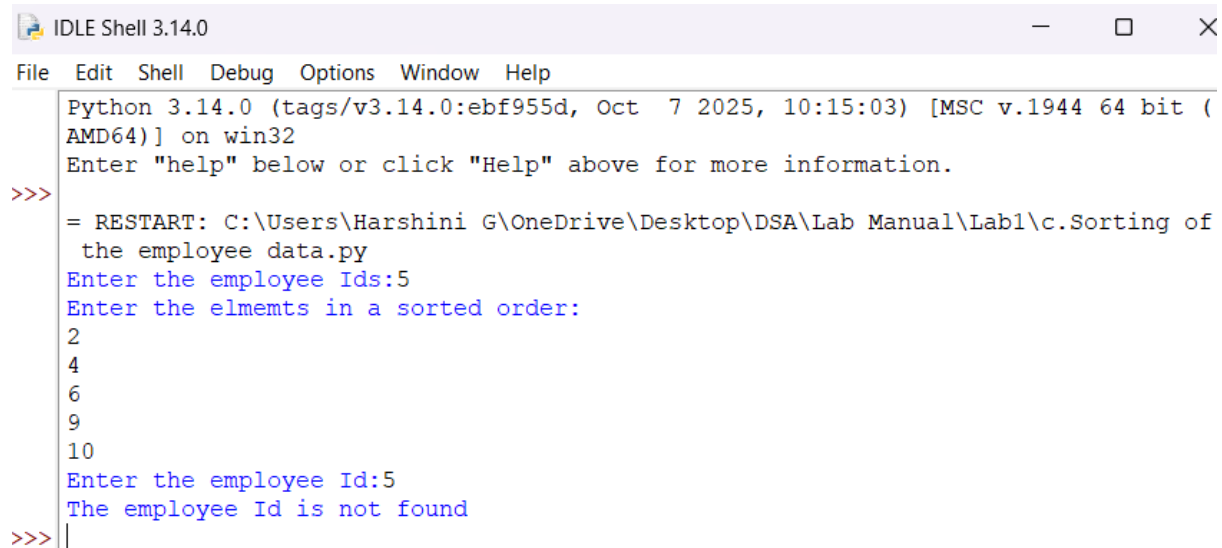
        high=mid-1

    return -1

n=int(input("Enter the employee Ids:"))
arr=[]
print("Enter the elmemts in a sorted order:")
for i in range(n):
    arr.append(int(input()))
key=int(input("Enter the employee Id:"))
result=binary_search(arr,key)
if result!=-1:
    print("The employee Id is found")
else:
    print("The employee Id is not found")

```

OUTPUT:



```

IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:\Users\Harshini G\OneDrive\Desktop\DSA\Lab Manual\Lab1\c.Sorting of
the employee data.py
Enter the employee Ids:5
Enter the elmemts in a sorted order:
2
4
6
9
10
Enter the employee Id:5
The employee Id is not found
>>> |

```

1d) **AIM:** In a logistics company, the no of possible ways to arrange n different parcels are calculated using the factorial formula($n!$). Develop a recursive program to calculate the factorial of a given number.

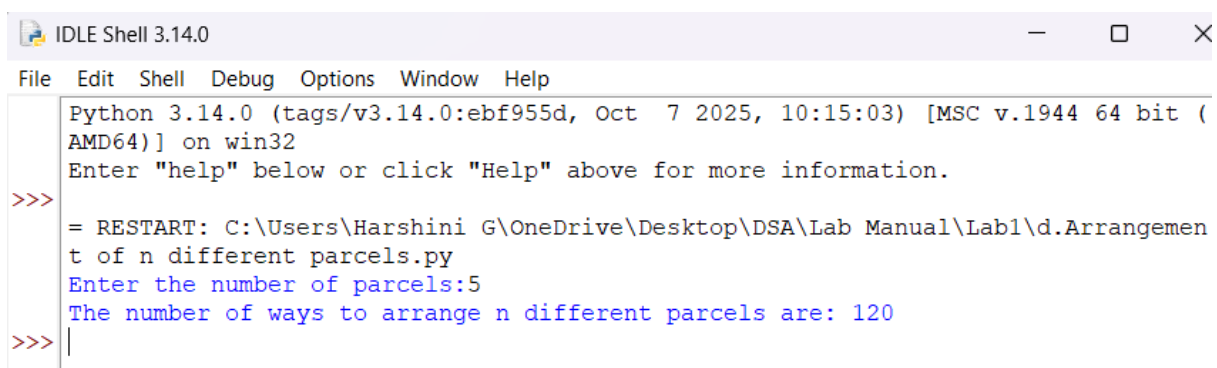
CODE:

```
def fact(n):
```

```
if n==0 or n==1:
    return 1
return fact(n-1)*n
```

```
n=int(input("Enter the number of parcels:"))
result=fact(n)
print("The number of ways to arrange n different parcels are:",result)
```

OUTPUT:



```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:\Users\Harshini G\OneDrive\Desktop\DSA\Lab Manual\Lab1\d.Arrangement of n different parcels.py
Enter the number of parcels:5
The number of ways to arrange n different parcels are: 120
>>> |
```

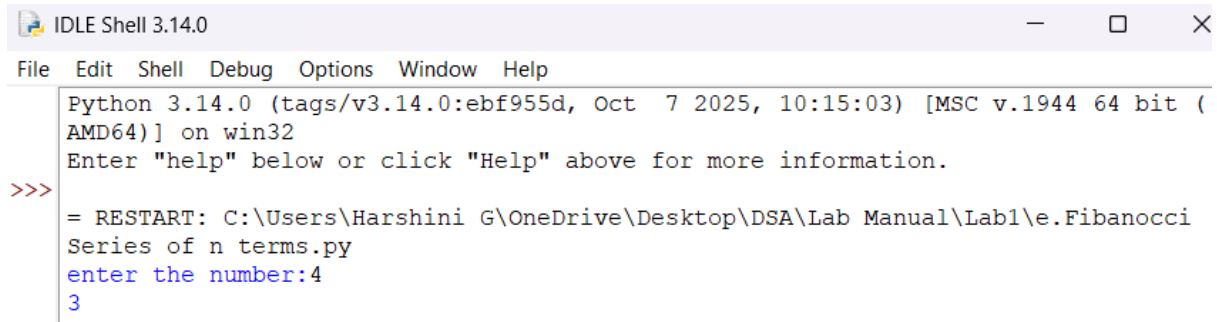
1e) **AIM:** Write a recursive python program to print the 1st 'n' terms of Fibonacci series.

CODE:

```
def fib(n):
    if n<=1:
        return n
    return fib(n-1)+fib(n-2)
```

```
n=int(input("enter the number:"))
result=fib(n)
print(result)
```

OUTPUT:

A screenshot of the IDLE Shell 3.14.0 window. The title bar says "IDLE Shell 3.14.0" with standard window controls. The menu bar includes "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The shell area shows the following text: "Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32", "Enter 'help' below or click 'Help' above for more information.", a prompt ">>>", and then the execution of a script: "= RESTART: C:\Users\Harshini G\OneDrive\Desktop\DSA\Lab Manual\Lab1\e.Fibonacci Series of n terms.py", "enter the number:4", and the output "3".

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\Harshini G\OneDrive\Desktop\DSA\Lab Manual\Lab1\e.Fibonacci Series of n terms.py
enter the number:4
3
```

WEEK-2

2a) **AIM:** Write a python program to implement linear search.

CODE:

```
def linear_search(arr,key):
    for i in range(len(arr)):
        if arr[i]==key:
            return i
    return -1

n=int(input("Enter the number of elements:"))
arr=[]
print("Enter the elements:")
for i in range(n):
    arr.append(int(input()))
key=int(input("Enter the elements to search:"))
result=linear_search(arr,key)
if result!=-1:
    print("Element is found at the index:",result)
else:
    print("Elemnet is not found at the index")
```

OUTPUT:

```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\Harshini G\OneDrive\Desktop\DSA\Lab Manual\Lab2\a.Linear search.py
Enter the number of elements:5
Enter the elements:
2
4
6
3
7
Enter the elements to search:3
Element is found at the index: 3
>>>
```

2b)

i) **AIM:** Write a python program to implement binary search with sorted array as an input.

CODE:

```
def binary_search(arr,key):
    low=0
    high=len(arr)-1
    while low<=high:
        mid=(low+high)//2
        if arr[mid]==key:
            return mid
        elif arr[mid]<key:
            low=mid+1
        else:
            high=mid-1
    return -1

n=int(input("Enter the number of elements:"))
arr=[]
print("Enter the Sorted elements:")
for i in range(n):
    arr.append(int(input()))
```



```

key=int(input("Enter the element to search:"))

result=binary_search(arr,key)

if result!=1:

    print("Element found at the index:",result)

else:

    print("Element is not found")

```

OUTPUT:

```

Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\Harshini G\OneDrive\Desktop\DSA\Lab Manual\Lab2\b1.Binary search of a sorted data.py
Enter the number of elements:5
Enter the Sorted elements:
2
5
6
7
8
Enter the element to search:7
Element found at the index: 3
>>>

```

ii) **AIM:** Write a python program to implement binary search without sorted array as an input.

CODE:

```

def binary_search(arr, key):

    low = 0

    high = len(arr) - 1

    while low <= high:

        mid = (low + high) // 2

        if arr[mid] == key:

            return mid

        elif arr[mid] < key:

            low = mid + 1

        else:

```

```

        high = mid - 1

    return -1

n = int(input("Enter the number of elements: "))

arr = []

print("Enter the elements:")

for i in range(n):

    arr.append(int(input()))

arr.sort()

print("Sorted array is:",arr)

key=int(input("Enter the element to search:"))

result=binary_search(arr,key)

if result!=-1:

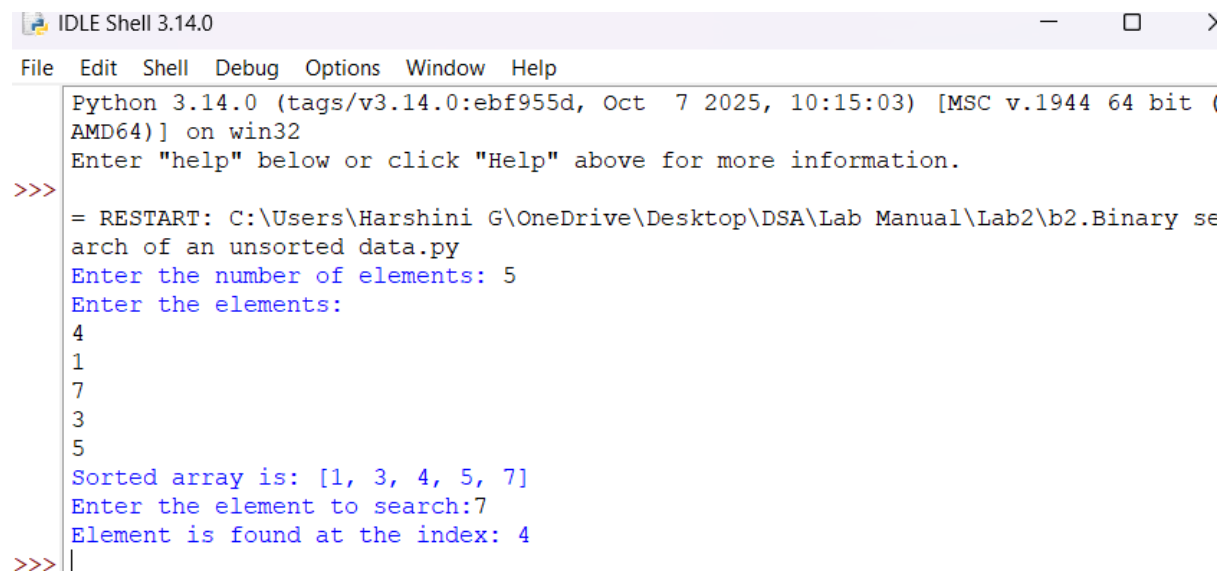
    print("Element is found at the index:",result)

else:

    print("Element is not found")

```

OUTPUT:



```

IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:\Users\Harshini G\OneDrive\Desktop\DSA\Lab Manual\Lab2\b2.Binary search of an unsorted data.py
Enter the number of elements: 5
Enter the elements:
4
1
7
3
5
Sorted array is: [1, 3, 4, 5, 7]
Enter the element to search:7
Element is found at the index: 4
>>>

```

WEEK-3

3a) **AIM:** s

CODE:

OUTPUT:

3b) **AIM:** s

CODE:

OUTPUT: